Simplified Nonlocal Network Based on Adaptive Projection Attention Method for Hyperspectral Image Classification

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***Abstract*— Nonlocal convolutional neural networks (CNNs) have difficulties in dealing with the imbalanced samples in hyperspectral images (HSIs) effectively, so the networks cannot achieve ideal experiment results. Therefore, this article proposes an adaptive projection attention-based simplified nonlocal neural network for HSI classification. First, the local information is calculated in horizontal and vertical directions. Then, the information is passed to a simplified nonlocal network to learn the global semantic information. The simplified nonlocal network can reduce information redundancy and improve classification accuracy at the same time. Second, the global semantic information is adaptively projected according to the spatial features and compressed using multiscale pooling layers. After that, the pooled results are reassigned channel weights through two fully connected layers and extended using multiscale pooling layers. Then, the extended features are concatenated with the global semantic information, which can alleviate the imbalanced sample existing in the dataset. Then, a simplified nonlocal approach is used to fuse shallow and deep information to improve the robustness and classification performance of the network. In this article, experiments of the proposed method are conducted on three widely used hyperspectral datasets compared with those of seven state-of-the-art algorithms, and satisfactory overall and average accuracies are achieved, demonstrating the effectiveness of the proposed algorithm.**

***Index Terms*— Attention mechanism, convolutional neural networks (CNNs), hyperspectral image (HSI) classification, nonlocal module.**

I. Introduction

With advanced satellite-based and airborne imaging spectrometers, hyperspectral imaging techniques have become increasingly sophisticated. A hyperspectral image (HSI) contains hundreds of consecutive spectral bands obtained by the spectral emission and reflection response of an object [1], [2]. The wealthy image spatial and spectral information has made HSI widely used in various applications, such as national defense, target detection [3], [4], agricultural production, and super-resolution [5]. Furthermore, HSI classification is an important research field in hyperspectral data analysis [6], [7]. HSI classification aims to determine the class each pixel belongs by analyzing the spatial and spectral information of HSI [8]. However, in the remote sensing field, only a small amount of labeled data are available, since collecting such labeled data is either expensive or time-demanding [9].

Most traditional HSI classification methods use supervised machine learning approaches. They perform spectral feature extraction and obtain the results by different classifiers, for example, support vector machine (SVM) [10], extreme learning machine (ELM) [11], and multinomial logistic regression (MLR) [12]. To solve the problem of redundant information, some studies focus more on dimensionality reduction, such as principal component analysis (PCA) [13], minimum noise fraction rotation (MNF Rotation) [14], and independent component analysis (ICA) [15]. Kernel PCA [13] and kernel MNF [16] are applied to deal with the nonlinear relationship between hyperspectral data. However, relying on spectral information alone does not maximize the knowledge of HSI, so numerous methods based on spatial–spectral features are applied to improve the accuracy of HSI classification. Among them are fixed domain methods, such as wavelet transform [17], Gabor filtering [18], and various morphological analysis methods based on morphological filtering [19]. There are also methods based upon the adaptive domain. Kang et al. [20] used edge-preserving filtering after classifying HSI by SVM to improve classification accuracy. Moreover, Ji et al. [21] constructed hypergraphs by interpixel relationships in a spectral space and an optimized hypergraph structure by learning boundary weights.

Over recent years, deep learning methods have attained miraculous achievements in HSI classification. Numerous deep learning algorithms have been widely used, such as convolutional neural network (CNN) [22], recurrent neural network (RNN) [23], generative adversarial network (GAN) [24],

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long short-term memory (LSTM) [25], and graph convolutional network (GCN) [26]. As a data-driven method, the CNN automatically extracts abstract semantic features by deploying hierarchical distributed structures. The CNN uses the receptive field to learn features and improves computational efficiency through weight sharing and the local connection of neurons. Zhu et al. [24] adopted a GAN to combine labels with noise to generate fake samples to mitigate the overfitting problem. At the same time, the corresponding 1-D CNN [27], 2-D CNN [28], and 3-D CNN [29] have been successfully applied in remote sensing data due to the better learning of spectral features, spatial features, and joint spatial–spectral features. Affected by the typical 3-D cube representation of HSI data, Li et al. [29] directly input the 3-D data block into CNN and efficiently extracted the spectral–spatial information.

However, due to the small training samples of the HSI, deep learning networks tend to be overfitted. Therefore, researchers have proposed several strategies to alleviate this dilemma, such as data augmentation (DA) [22], transfer learning (TL) [30], and unsupervised/semi-supervised feature learning [31], [32]. DA is a natural approach to coping with few training samples. It mainly generates fictitious samples to balance the number of samples by two strategies. One strategy is transformation-based sample generation; i.e., the existing samples are utilized by a transformation function with weighted Gaussian noise to generate new samples of the same class. For example, Haut et al. [33] obtained new samples by developing occluded boxes of random size and setting the preset values of the pixels inside the boxes. The other strategy is combination-based sample generation, where a new sample is generated by a convex combination of two existing samples. The convex combination basically ensures that the new sample is the same class. Kang et al. [34] used the convex combination of two samples in the same class to construct a virtual sample, ensuring that the new sample belonged to the same class.

However, most CNN algorithms mentioned above use fixed-size image blocks as input, often bringing two problems. First, the computational consumption becomes unavoidably higher as a consequence of repeated computations of overlapping parts. Second, the use of predefined window sizes [35] causes rare utilization of the global semantic information, thus reducing the learning potentiality of the network. Liang et al. [36] used multiscale image blocks to obtain long-range semantic information to solve the above problems. However, large-scale convolutional kernels can blur the image's intricate parts, resulting in less distinct features.

Recently, neural networks based on nonlocal modules have emerged as antidotes for directly extracting distant image dependencies [37]. Buades et al. [38] first used the nonlocal mean method for image noise reduction. This method computes the mean values between pixels of the whole image instead of local fields, thus increasing the robustness of image filtering and helping eliminate noise in the image. Wang et al. [39] introduced the nonlocal idea into CNN to extract global semantic information based on a self-attentive mechanism. However, for some larger images, computing the response of each pixel to the overall image will undoubtedly consume a lot of computational resources and time. A fully convolutional network with an efficient nonlocal module (ENL-FCN) was proposed based on a fully CNN [40]. The ENL-FCN reduces the time complexity by calculating the response between a pixel and its crisscross peers. However, this method does not involve the problem of imbalanced samples in HSI. Moreover, a dire information redundancy situation exists in obtaining the full long-range dependency image, preventing the network from learning discriminative features expeditiously. Therefore, this article uses a simplified nonlocal (SNL) method to learn the long-range semantic information and reduces the information redundancy. Besides, Cao et al. [41] found that the query branch was unnecessary in nonlocal operations. They combined the simplified nonlocal and attention module to get better results. They removed the query branch and compressed the channel information to achieve the simplified nonlocal module. However, compressing the whole image into a single value is not an efficient way to deal with imbalanced samples. Compared with [41], this article focuses more on the minor samples while deploying the attention module to relieve the sample imbalance problem. The loss function is also adjusted to learn more information about the minor samples. At the same time, this article fuses the outputs of different layers to get a more accurate classification.

To relieve the sample imbalance problem in HSI and use the long-range semantic information efficiently, this article proposes a simplified nonlocal neural network based on the adaptive projection attention method (APAM). This article removes the query branch to use the long-range semantic information efficiently. Adaptive pooling layers are adopted to regulate the sizes of other branches to solve the calculation problem after removing the query branch. The long-range semantic information is obtained after the multiplication of the other two branches. Then, the long-range semantic information is adaptively projected based on spatial features to alleviate the problem of imbalanced samples and learn the critical information of the minor samples effectively. Different sizes of effective information are extracted through multiscale pooling layers. After that, two fully connected layers are exploited to reassign channel weights. Before concatenating with global semantic information, the reassigned result is passed to a pooling layer to extend the spatial size of the image. Next, several convolutional layers are exploited to learn more abstract features. Finally, to effectively utilize the information from different layers, the shallow information is fused with the deep one through a nonlocal fusion network to improve the robustness and classification performance of the network. The main contributions of the proposed APSNL are as follows.

1) An SNL method is used to learn long-range semantic information efficiently. The query branch is removed in the method, since it has little impact on the nonlocal correlation. At the same time, the adaptive pooling layers are adopted to regulate the sizes of other branches.

2) An improved attention method focuses more on spatial information to reinforce the network's ability to detect small-scale changes. Multiscale spatial information derived by adaptive projection helps recognize the minor samples, improving

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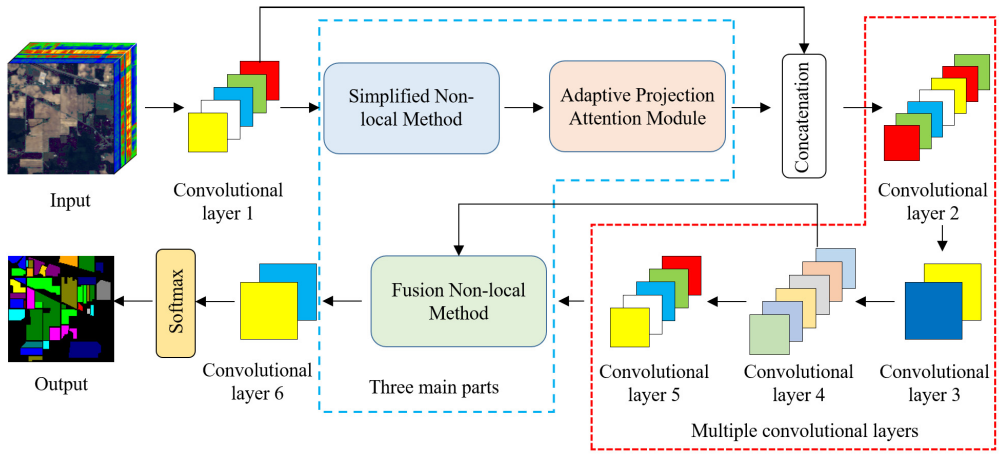


Fig. 1. Overall framework of the proposed ASPNL.

the network's ability to deal with the sample imbalance problem.

3) A fusion nonlocal module is performed to fuse features of different layers. The module learns how the output of shallow and deep convolutional layers is related in terms of long-range dependencies. As a result, the network may learn to produce accurate classification results that are both complete and holistic.

The rest of this article proceeds in this manner. Section II describes the proposed APSNL for HSI classification. Section III documents the experimental results and displays the analysis on three datasets, and Section IV concludes this article.

II. Proposed Method

To address the sample imbalance problem of HSI data, this article proposes a simplified nonlocal network based on the adaptive projection attention module. Fig. 1 exhibits the elementary procedure of the proposed ASPNL. In Fig. 1, the convolutional layers 1–6 are six convolutional layers, and the concatenation operation is applied to reuse the outputs of convolutional layer 1 to get better classification performance. As shown in Fig. 1, the proposed ASPNL takes the 3-D HSI as an input. The image data are first processed using normalization to prevent the gradient explosion phenomenon caused by too many parameters. The feature extracted initially by convolutional layer 1 is passed to the SNL method to learn the global semantic information. The APAM can detect spatial changes in the image to recognize imbalanced samples. Then, multiple convolutional layers, composed of convolutional layers 2–5, are used to get different scales of receptive fields to enhance the nonlinear representation of the network. Finally, the fusion nonlocal (FNL) method is used to fuse the results of convolutional layers 4 and 5 to optimize the performance of ASPNL. Sections II-A–II-C will detail the proposed elements of the network, SNL, APAM, and FNL.

A. Simplified Nonlocal Method

Nonlocal methods are deployed to obtain long-range semantic information to enhance learning performance. During the deployment, it is found that the query branch can be removed, since it contributes little to the output. However, removing the query branch brings computational problems of image size. Based on the above facts, this article proposes the SNL method. By three 1×1 convolutional layers, the original nonlocal module first obtains three branches, query, key, and value. Then, the whole image's long-range semantic information is formulated as follows:

$$y = x + F_V(x) \times \left(F_Q(x) \times F_K(x)\right) \tag{1}$$

where x and y are the input and output, and $F_Q(x)$, $F_K(x)$, and $F_V(x)$ are the results obtained by the query, key, and value branches. However, obtaining the similarity matrix by multiplying the query branch and the key branch consumes enormous computational resources. For a single image of $R^{H\times W}$, the size of the similarity graph is $R^{\mathrm{HW}\times\mathrm{HW}}$. Meanwhile, repeated computation occurs in earning the similarity graph, thus leading to information redundancy. Cao et al. [41] found that removing the query branch did not impact much on the final classification result. They solved the calculation problem by compressing the dimensions of the image in the key branch. The reduction of the query branch means that the advantage of inputting an arbitrary size image, which the original nonlocal network has, is no longer available. For the image input as $X \in R^{C\times H\times W}$, the size of the data becomes $R^{C\times\mathrm{HW}}$ after 1×1 convolution layers of the query, key, and value branches. When results of the query branch and the key branch are multiplied to obtain the similarity matrix, the size of the data becomes $R^{\mathrm{HW}\times\mathrm{HW}}$. Then, the size of the data by multiplying the similarity matrix and the value branch is $R^{C\times\mathrm{HW}}$. Finally, after 1×1 convolution layers, the size of data finally becomes $R^{C\times H\times W}$. In the calculation process, if the query branch is removed, the result of multiplying the key branch and the value branch will become data of size $R^{\mathrm{HW}\times\mathrm{HW}}$ or $R^{C\times C}$, and such result cannot become the same size as the input. Therefore, this article proposes an SNL method based on two branches for learning long-range contextual information. The SNL reduces computational consumption and information redundancy by removing the query branch. Also, it improves the classification accuracy by obtaining the responses between

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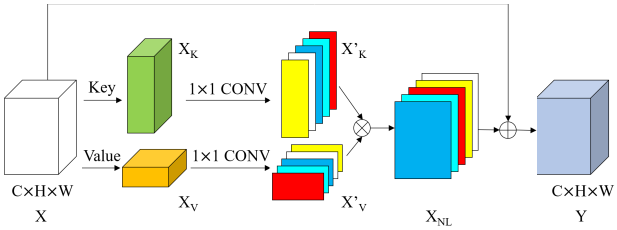


Fig. 2. Architecture of the SNL method.

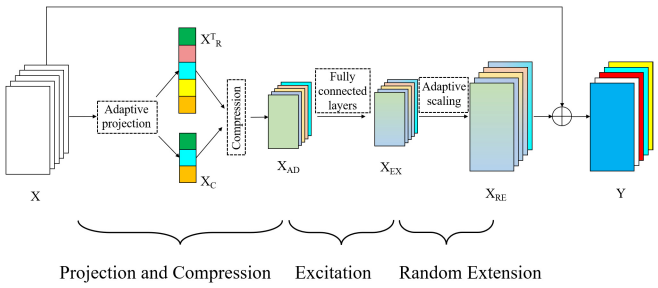


Fig. 3. Framework of the proposed APAM.

different pixels. Fig. 2 displays the elementary framework of the SNL.

Given an HSI dataset denoted as $X_{\mathrm{SNL}} \in R^{C\times H\times W}$, where C refers to the amounts of spectral bands, H and W represent the height and the width of the spatial dimension, respectively. As shown in Fig. 2, SNL obtains X_K and X_V through adaptive pooling in the key branch and the value branch

$$X_K = F_1(X_{\mathrm{SNL}}), \quad X_V = F_2(X_{\mathrm{SNL}}) \tag{2}$$

where F_1 and F_2 are two different adaptive pooling functions. Also, the output size of F_1 is $R^{C\times H\times B}$, and the output size of F_2 is $R^{C\times B\times W}$. Also, B is the changed size by the adaptive pooling layers. After the dimension changes, the results of the two branches are passed to the convolutional layers to obtain X'_K and X'_V. As shown in Fig. 2, the layers do not adjust the image size but only the number of channels through the 1×1 convolutional kernel sizes. Then, X'_K and X'_V are multiplied to obtain the long-distance semantic information X_{NL}

$$X_{\mathrm{NL}} = \frac{1}{N} X'_K \times X'_V \tag{3}$$

where N is a normalization factor. The change of image dimension ensures that the image size of long-distance semantic information is the same as that of input. By concatenating the input X_{SNL} and X_{NL}, the output Y_{SNL} can be described as follows:

$$Y_{\mathrm{SNL}} = \mathrm{cat}(X_{\mathrm{SNL}}, X_{\mathrm{NL}}). \tag{4}$$

B. Attention Method Based on Adaptive Projection

Most CNNs pay attention to learning features using convolutional layers in spatial directions. The processing of channel dimensions is mostly simple grouping to reduce the amount of calculation. The lack of learning information of channels restricts the performance of the networks. The squeeze-excitation network (SENet) is originally proposed by Hu et al. [42]. Hu et al. innovatively conducted interchannel learning, so that the network could better understand the importance of different channels. However, compressing the whole image into a single value is not an efficient way to learn the partial spatial features. Thus, the APAM is proposed to fully use the spatial information to enhance change detection. The framework of the proposed APAM is delineated in Fig. 3.

The whole procedure of the proposed APAM is separated into three portions. In the first step, the output results of the previous network are adaptively projected according to the spatial information and compressed using adaptive pooling layers. The compressed results can learn the importance of channels through two fully connected layers in the second step. In the third step, the image is expanded according to the adaptive pooling layers to retain spatial information. Each part will be depicted in detail below.

1) Adaptive Projection and Compression: The SENet's method of compressing the whole image into a single point ignores the spatial information of the image. So, it is not easy to learn the information when facing the HSI with imbalanced samples. Therefore, this article proposes an adaptive projection and compression process to strengthen the feature learning of the minor samples by taking the spatial information into account. The projection and compression part in Fig. 3 shows that the input $Y_{\mathrm{SNL}} \in R^{C_R\times H\times W}$ uses spatial information to adaptively generate the projection interval in the horizontal and vertical directions, respectively. Then, the row features $X_R \in R^{C_R\times H\times 1}$ and column features $X_C \in R^{C_R\times 1\times W}$ are acquired according to the projection interval. Because H and W of HSI data are often not equal, there is a case of repeated selection in this procession, which can better learn information of the minor samples. Then, X_R and X_C are passed to multiscale adaptive average pooling layers to extract and compress the spatial essential information to obtain X^R_{AD} and X^C_{AD}

$$\begin{aligned} X^R_{\mathrm{AD}} &= \mathrm{cat}(A_1(X_R), A_2(X_R), \ldots, A_n(X_R)) \\ X^C_{\mathrm{AD}} &= \mathrm{cat}(A_1(X_C), A_2(X_C), \ldots, A_n(X_C)) \end{aligned} \tag{5}$$

where A_i represents the adaptive pooling functions of different scales, $i = 1, 2, \ldots, n$. Also, n is the number of adaptive pooling layers. The sizes of X^R_{AD} and X^C_{AD} are $R^{C\times N_R\times 1}$ and $R^{C\times 1\times N_R}$, where N_R and N_C are the sum of spatial dimensions of pooling results corresponding to X^R_{AD} and X^C_{AD}. Multiscale pooling layers use different sizes of receptive fields to obtain features. On the one hand, richer features are acquired, so that the proposed APAM can learn from imbalanced samples effectively. On the other hand, after taking appropriate window sizes, multiscale pooling can reduce the time consumption.

2) Excitation: $X_{\mathrm{AD}} \in R^{C\times N_R\times N_C}$ is obtained by adding compressed row features X^R_{AD} and column features X^C_{AD}. HSI, composed of hundreds of spectral bands, has rich channel information. In this article, the importance of channels is learned through two fully connected layers. X_{AD} is passed to two fully connected layers to obtain $X_{\mathrm{EX}} \in R^{C\times N_R\times N_C}$

$$X_{\mathrm{EX}} = F_{\mathrm{EX}}(X_{\mathrm{AD}}) = \delta_2(W_2\delta_1(W_1 X_{\mathrm{AD}})) \tag{6}$$

where $W_1 \in R^{(C/r)\times C}$ and $W_2 \in R^{C\times(C/r)}$ are two fully connected layers, which introduce a scaling factor r to reduce the number of parameters. δ_1 is the activation function rectified

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linear unit function (ReLU) after the first fully connected layer, and δ_2 is the activation function sigmoid after the second fully connected layer.

3) Random Extension: In the compression procedure, information loss is inevitable. Therefore, to alleviate this phenomenon and maintain the consistency of image space size, this article proposes a random extension method, using the excitation result X_{EX} to obtain the extension result $X_{\mathrm{RE}} \in R^{C\times H\times W}$. Due to the different pooling scales in X_{EX}, if the results of multiple scales are taken as a whole, it is easy to cause local distortion of information in the restored image. Therefore, it is necessary to obtain the restored image in each pooling result based on its pooling scale. Since X_{EX} is obtained by adding X^R_{AD} and X^C_{AD} in different dimensions, the image restoration needs to be performed in both dimensions. First, the horizontal information X^R_{EX} and the vertical information X^C_{EX} of X_{EX} are obtained as follows:

$$X^R_{\mathrm{EX}} = \frac{1}{N_R}\sum_{j=1}^{N_R} X_{\mathrm{EX}}(:, j, :)$$
$$X^C_{\mathrm{EX}} = \frac{1}{N_C}\sum_{k=1}^{N_C} X_{\mathrm{EX}}(:, :, k) \tag{7}$$

where N_R and N_C are the corresponding spatial information of X_{EX}. Then, pooling is operated on each scale to recover spatial information

$$\begin{aligned} X^R_{\mathrm{RE}i} &= A'_i\left(X^R_{\mathrm{EX}i}\right) \\ &= A'_i\left(X^R_{\mathrm{EX}}\left[\sum_{n=0}^{i-1}\mathrm{length}(A_{i-1}), \sum_{n=0}^{i}\mathrm{length}(A_i)\right]\right) \\ X^C_{\mathrm{RE}i} &= A'_i\left(X^C_{\mathrm{EX}i}\right) \\ &= A'_i\left(X^C_{\mathrm{EX}}\left[\sum_{n=0}^{i-1}\mathrm{length}(A_{i-1}), \sum_{n=0}^{i}\mathrm{length}(A_i)\right]\right) \end{aligned} \tag{8}$$

where A'_i represents the inverse operation of the pooling function corresponding to (5). At the same time, because X_{AD} is the sum of the concatenation of multiple pooling results, the pooling results on each scale can be determined by calculating the sum of adjacent pooling scales. Then, the restored image X_{RE} can be obtained by combining the corresponding restored images in the horizontal and vertical directions, that is,

$$\begin{aligned} X_{\mathrm{RE}} &= \frac{1}{n}\sum_{i=1}^{n} X_{\mathrm{RE}i} \\ &= \frac{1}{n}\sum_{i=1}^{n}\left(X^R_{\mathrm{RE}i} + X^C_{\mathrm{RE}i}\right) \end{aligned} \tag{9}$$

where $X_{\mathrm{RE}i}$ is the recovered image obtained from the dialog results at different scales.

Since the results of multiscale pooling in X_{EX} must be processed separately, it is undeniable that this step is computationally intensive and time-consuming. Still, X_{RE} retains the information of multiple scales and has a more outstanding contribution to learning the features of the minor samples. Finally, X_{RE} and Y_{SNL} are multiplied pixel-wise to obtain the output result $Y_{\mathrm{APAM}} \in R^{C_R\times H\times W}$

$$Y_{\mathrm{APAM}}(i, j, k) = X_{\mathrm{RE}}(i, j, k)Y_{\mathrm{SNL}}(i, j, k) \tag{10}$$

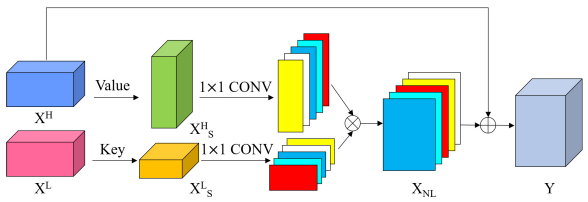


Fig. 4. Framework diagram of the FNL method.

where $i = 0, 1, \ldots, C-1$; $j = 0, 1, \ldots, H-1$; $k = 0, 1, \ldots, W-1$.

Compared with the SNL, random extension aims to recover the spatial information of the adaptive compression and projection results after the channel weight changes to alleviate the information loss caused by the compression process. Random extension leads to relatively significant time consumption because of the calculation on each pooling scale. Still, due to the adaptive pooling on the different scales as far as possible to restore the spatial information of the image, the accuracy decline caused by information loss is effectively reduced.

C. Fusion Nonlocal Method

To achieve multifeature fusion and get better classification, this article proposes a fusion SNL method and FNL. The information derived by the deep learning methods alters due to the different compositions of the network. When the network is deeper, the level of semantic information extracted is higher, but the target's location is fuzzier. Moreover, the overfitting phenomenon will become more severe with the increase in network layers. To make the classification result more accurate and reduce the occurrence of the network overfitting, multifeature fusion is necessary. Most of the existing feature fusion methods focus on spatial and channel dimensions. They cannot learn the response between the output results of different layers. In the contrast, the FNL uses the SNL method to fuse the features of different layers to get better classification. The concise procedure of the proposed FNL is demonstrated in Fig. 4.

In Fig. 4, to achieve multifeature fusion, the FNL takes the results of convolutional layers 4 and 5 as an input, where $X^L \in R^{C_L\times H\times W}$ and $X^H \in R^{C_H\times H\times W}$ represent low-level features from convolutional layer 4 and high-level features from convolutional layer 5, respectively, and H and W represent the height and the width of the spatial dimension, while C_L and C_H represent the channels of low-level features and high-lever features. To obtain sufficient local information of these two features, adaptive pooling layers are applied to acquire X^L_S and X^H_S, which are calculated as follows:

$$X^L_S = F_1\left(X^L\right), \quad X^H_S = F_2\left(X^H\right) \tag{11}$$

where F_1 and F_2 are two different adaptive pooling functions. Since the pooling process takes place in the spatial dimension, the number of channels of the sampling results is not changed. Therefore, the sampling results are passed through a 1×1 convolutional layer to adjust the number of channels of two branches while keeping the spatial dimension consistent. The long-distance dependency of low-level features

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TABLE I
PARAMETERS OF SIX CONVOLUTIONAL LAYERS

Layers	Kernel sizes	Strides	Group	Activation function
1	5×5	2×2	Yes	BN + sigmoid
2	5×5	2×2	Yes	BN + sigmoid
3	5×5	2×2	Yes	BN + sigmoid
4	3×3	1×1	Yes	BN + sigmoid
5	3×3	1×1	Yes	BN + sigmoid
6	1×1	0×0	No	No

and high-lever features is calculated by the multiplication operation. In other words, X_{NL} can be represented by

$$X_{\mathrm{NL}} = \mathrm{act}\left(F_K\left(X_S^L\right)\right) \times F_V\left(X_S^H\right) \tag{12}$$

where act is the activation function SoftMax, and F_K and F_V are the convolutional functions of the key branch and value branch, respectively. X^H has deeper semantic features than X^L, so the output Y can be calculated as follows:

$$Y = \mathrm{cat}\left(X^H, X_{\mathrm{NL}}\right). \tag{13}$$

The proposed FNL takes the outputs of different layers as an input. It takes advantage of the long-distance semantic information of different layers to enhance the network's feature learning ability and nonlinear ability.

D. Convolutional Layers Parameters and Loss Function

There are six different convolutional layers in the proposed ASPNL. The first five convolutional layers adopt group convolution, and there is a sigmoid activation function behind these five convolution layers as a nonlinear layer. Table I details the parameter settings of these convolutional layers.

The original cross-entropy function performs poorly when facing the sample imbalance problem of HSI. It is impracticable to scientifically learn from the minor samples in the training process. A cross-entropy function based on mean frequency balance is proposed in [43]. Considering that the number of pixels to be classified in HSI sometimes only occupies a part of the whole image, this article modifies the weight W in [43] to

$$W = \frac{M}{F} = \frac{M \times \mathrm{pixels}}{S} \tag{14}$$

where M represents the average value of a category, and F represents the category frequency. The final weighted cross entropy can be described as follows:

$$\mathrm{Loss}^{\mathrm{CE}}(y, \hat{y}) = -\sum_{i=1}^{n} W_i y_i \log(\hat{y}_i) \tag{15}$$

where y and $\hat{y}$ are actual and predicted class labels, respectively, and n is the number of categories.

E. Computational Complexity

The proposed algorithm is mainly composed of SNL, APAM, and FNL. The SNL method and the FNL method have the same structure, so the SNL method and the APAM method are the main ones that affect the algorithm's complexity. The SNL uses the adaptive pooling layer to obtain local information and simplify the nonlocal operations. In the original nonlocal operation, for the image as $R^{C\times H\times W}$, obtaining the similarity matrix requires $C \times \mathrm{HW} \times \mathrm{HW}$ times of multiplication. While in SNL, obtaining the similarity matrix requires $C \times \mathrm{HW} \times B$ times of multiplication. Since B is much smaller than HW, the complexity of the SNL is reduced compared with the original nonlocal operation. In APAM, there are $C \times (H \times N_R + W \times N_C)$ times of multiplication in adaptive projection progress and random extension progress, where N_R and N_C are the sums of spatial dimensions of pooling results in Section II-B. Compared with traditional attention mechanisms, such as SENet, APAM enriches the learning of spatial information through adaptive projection and compression in the horizontal and vertical directions. Then, two fully connected layers are used to adjust the channel weight. To effectively combine the modified channel weight with the input information, the random extension method is used to adjust the image size to obtain the final output. Many changes in the image size in the APAM method will lead to an increase in computational consumption. However, because of significantly considering the spatial information, the APAM can effectively learn the information of the minor samples and alleviate the sample imbalance problem in HSI.

The basic steps of the algorithm proposed in this article are shown in Algorithm 1.

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Datasets Description

In this article, classification experiments are carried out on three widely used datasets to verify the performance of the proposed ASPNL. They are the Indian Pines (IP) dataset, the University of Pavia (UP) dataset, and the Kennedy Space Center (KSC) dataset. The IP dataset faces a serious category imbalance problem. Also, for the KSC and UP datasets, some samples still face imbalance problems when the training ratio is less. The specific information of the datasets is as follows.

1) Indian Pines: The dataset is the image of the Indian Pines area in northwest Indiana, USA, obtained by the airborne visible/infrared imaging spectrometer (AVIRIS). The data contain 145 × 145 pixels, including 220 spectral reflection bands. After removing bands disturbed by water surface absorption and ozone, the final test dataset contains 200 bands. This dataset includes 16 features with 10 249 marked pixels. In our experiment, 10% of samples from each class are randomly selected as training sets, while 90% are testing sets. Table II shows the corresponding colors, land cover information, and sample number of specific classes.

2) Kennedy Space Center: This image is derived by National Aeronautics and Space Administration (NASA) at the Kennedy Space Center using AVIRIS. The test dataset image consists of 512 × 614 pixels, including 224 spectral

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Algorithm 1 ASPNL: Simplified Nonlocal Network Based on the Adaptive Projection Attention Method

Input: original HSI data X; maximum number of iterations I, number of adaptive pooling layers in APAM N, adaptive pooling operation A and its inverse operation A';

Output: Classification result Y;

X_{SNL} is obtained by standardizing X;
while the number of iterations $< I$ **do**
  Employ the adaptive pooling layers in the horizontal and vertical directions of X_{SNL} to get X_{BH} and X_{BW}, and the long-range dependency information Y_{SNL} is obtained by using a simplified nonlocal network;
  Take the average in the horizontal and vertical directions of Y_{SNL} to get X_R and X_C;
  $X_{AD}^R \leftarrow []$, $X_{AD}^C \leftarrow []$;
  for $i = 1$ to N **do**
    $X_{AD}^R \leftarrow cat(A_i(X_R), X_{AD}^R)$;
    $X_{AD}^C \leftarrow cat(A_i(X_C), X_{AD}^C)$;
  end for
  Broadcast add X_{AD}^R and X_{AD}^C to get X_{AD};
  Adjust channel weights of X_{AD} through the full connection layer to get X_{EX};
  for $i = 1$ to N **do**
    $X_{EXi} \leftarrow X_{EX}[A_i, sum(A_i)]$;
    $X_{REi} \leftarrow A_i'(X_{EXi})$;
  end for
  Calculate the mean value of the results of different pool layers to obtain X_{RE};
  Multiply X_{RE} and Y_{SNL} to get Y_{APAM};
  Use multiple convolution layers to obtain shallow feature X^L and deep feature X^H;
  Pass X^L and X^H to FNL to get the final classification result Y.
end while

reflection bands. The final test dataset contains 176 bands after taking away 48 due to water surface absorption or low signal-to-noise ratio. The test dataset includes 5202 marked pixels and 13 kinds of land cover objects, including shrubs, swamps, salt marshes, and other ground objects; 5% of samples from each class are randomly selected as training sets, while the remaining 95% are testing sets. Table III shows the corresponding colors, land cover information, and sample number of specific classes.

3) University of Pavia: This dataset is collected at Pavia University, Italy. After removing 12 noisy bands, the final test dataset contains 103 bands with 610 × 340 pixels. A total of 42 776 marked pixels of nine kinds of ground objects are included in the UP dataset. In our experiment, 1% of samples from each class are randomly selected as training sets, while 99% are testing sets. Table IV lists the corresponding colors, information, and sample number of specific classes.

In this article, the overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) are used to assess the results. OA refers to the proportion of correctly classified samples with all samples in the test process. The average percentage of accurately classified samples in each class is AA, stressing each class's classification accuracy. Kappa is calculated by a confusion matrix, which can effectively appraise the performance of methods facing imbalanced samples.

TABLE II
LAND COVER CLASSES AND NUMBERS OF SAMPLES FOR TRAINING AND TESTING ON THE IP DATASET

Color	Land cover class	Sample	Training	Testing
	Background	10776		
	Alfalfa	46	5	41
	Corn notill	1428	143	1285
	Corn min	830	83	747
	Corn	237	24	213
	Grass/Pasture	483	49	434
	Grass/Trees	730	73	657
	Grass/pasture-mowed	28	3	25
	Hay windrowed	478	48	430
	Oats	20	2	18
	Soybeans notill	972	98	874
	Soybeans min	2455	246	2209
	Soybeans clean	593	60	533
	Wheat	205	21	184
	Woods	1265	127	1138
	Bldg Grass Tree Drives	386	39	347
	Stone steel towers	93	10	83
	Total	21025	1031	9218

TABLE III
LAND COVER CLASSES AND NUMBERS OF SAMPLES FOR TRAINING AND TESTING ON THE KSC DATASET

Color	Land cover class	Sample	Training	Testing
	Background	309157		
	Scrub	761	39	722
	Willow-swamp	243	13	230
	CP-hammock	256	13	243
	Slash-pine	252	13	239
	Oak/Broadleaf	161	9	152
	Hardwood	229	12	217
	Swap	105	6	99
	Graminoid-marsh	431	22	409
	Spartina-marsh	520	26	494
	Cattail-marsh	404	21	383
	Salt-marsh	419	21	398
	Mud-flats	503	26	477
	Water	927	47	880
	Total	314368	268	4943

B. Experimental Settings

To verify the effectiveness of the proposed method, this article compares ASPNL with seven deep learning methods, including residual network (ResNet) [44], ContextNet [45], spectral–spatial residual network (SSRN) [46], deep pyramidal ResNet (DPyResNet) [47], attention-based adaptive spectral–spatial kernel ResNet (A^2S^2K-ResNet) [48], Nonlocal [40], and ENL-FCN [40] methods. Fuse-Adam is investigated to improve the proposed ASPNL in this article. The weight decay

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TABLE IV
LAND COVER CLASSES AND NUMBERS OF SAMPLES FOR TRAINING AND TESTING ON THE UP DATASET

Color	Land cover class	Sample	Training	Testing
	Background	164624		
	Asphalt	6631	67	6564
	Meadows	18649	187	18462
	Gravel	2099	21	2078
	Trees	3064	31	3033
	Painted metal sheets	1345	14	1331
	Bare Soil	5029	51	4978
	Bitumen	1330	14	1316
	Self-Blocking Bricks	3682	37	3645
	Shadows	947	10	937
	Total	207400	432	42344

in the experiments is set to $5e-6$, and the initial learning rate is $5e-5$ and decay according to the following equation:

$$\text{learning}_{\text{rate}} = \left(1 - \frac{\text{iter}}{\text{total}}\right)^{0.8} \tag{16}$$

where total is set to 2000 on three datasets. The network is implemented under PyTorch framework with an Intel i7-9750 CPU and an NVIDIA RTX2060 GPU. All experiments are repeated ten times to prevent chance occurrence.

C. Classification Results and Analysis

The experimental results of seven state-of-the-art deep learning methods and ASPNL on the IP dataset are given in Table V. From Table V, it is clear that the ASPNL outperforms the other methods in terms of OA, AA, and Kappa. The ASPNL achieves the best results in each class accuracy on 14 classes except Corn notill class and Soybeans min class. A^2S^2K-ResNet and ENL-FCN, respectively, obtain these two classes' best results. As seen in Tables II and V, the proposed algorithm achieves 100%, 100%, and 96.67% on the Alfalfa class, Grass/pasture-mowed class, and Oats class with five, three, and two training samples, respectively, which strongly shows that the proposed method can effectively handle imbalanced samples. Fig. 5 visualizes the classification results of all the methods experimented on the IP dataset. The classification results plots for ResNet, ContextNet, and DPyResNet, shown in Fig. 5(c), (d), and (f), respectively, have a more significant noise, because they lack long-distance information. Combined with the specific class information given in Table II, it can be seen that the SSRN and A^2S^2K-ResNet methods in Fig. 5(e) and (g) have classification errors when facing the Oats class on the left-hand side of the figure, because there are only two training samples. The classification results of both Nonlocal and ENL-FCN networks, as shown in Fig. 5(h) and (i), also have classification errors when facing the Oats class. In contrast, the proposed method classifies all of them correctly when facing the Oats class.

As for the KSC dataset, Table VI shows that the ASPNL outperforms other methods on OA, AA, and Kappa. The results in Table VI show that the ASPNL achieves the highest OA, AA, and Kappa, that is, 99.47%, 99.56%, and 99.42% specifically. Among the accuracy of each class, the proposed method achieves the best results in nine classes out of 13. The best results on the Scrub and Mud-flats are achieved by the SSRN method. Both the Nonlocal and ENL-FCN methods achieve 100% accuracy on the cabbage palm (CP)-hammock class and Spartina-marsh class. In Hardwood class, Swamp class, and Graminoid marsh class, the proposed method reaches 100% according to the experiment. Furthermore, the training samples in Hardwood class and Swamp class are only six and 12, again demonstrating the outstanding performance of the proposed method in the face of fewer training samples. Fig. 6 visualizes the classification results of all methods in the experiment on the KSC dataset. Because the KSC dataset samples are relatively balanced and independent, we select the case of conflict location and mark it with a white box in the figure. It is evident in Fig. 6(c) and (f) that the classification results of ResNet and DPyResNet have considerable misclassification in the white box, and the data should be classified as the slash pine. In Fig. 6(d) and (g), the Contextnet method and the A^2S^2K-ResNet method also have this problem, but the number of misclassification is smaller than the previous ones. The SSRN method and the A^2S^2K-ResNet method have misclassification on the borders of different classes in Fig. 6(e) and (g). Because these approaches employ fixed windows for convolution computation, long-distance information is given less emphasis. As shown in Fig. 6(g), the nonlocal method misclassifies the box's top-left corner, because the number of samples in the area is small. The proposed method visually produces the best result of the eight methods due to successful ASPNL, as shown in Fig. 6(i).

On the UP dataset, it is known from Table VII that the proposed method outperforms the comparison method regarding OA, AA, and Kappa. In the experiments on the UP dataset, the proposed method attains 99.28% OA and achieves the best results on six of nine classes. The result of the ASPNL method differs from the best results on the Asphalt class, Meadows class, and Self-Blocking Bricks class by 0.22%, 0.06%, and 1.00%, respectively. The ENL-FCN method achieves the best results on the Asphalt class and the Self-Blocking Bricks class, and the best result on the Meadows class is achieved by the Nonlocal method. In contrast, the OA of ResNet, ContextNet, and DPyResNet is lower. Fig. 7 displays the classification results from all the methods on the UP dataset. In Fig. 7(c), (d), and (f), the classification results of ResNet, ContextNet, and DPyResNet have significant noise accompanied by large misclassifications, because these methods cannot learn features correctly. They also do not cope well with the less balanced samples in HSI. Due to the lack of long-range information, SSRN generates an inaccurate result with misclassification on the boundaries of different classes in Fig. 7(e). The classification results of A^2S^2K-ResNet in Fig. 7(g) are suitable when facing the border and narrow sample distribution, but misclassification occurs when facing the Bare Soil class. As shown in Fig. 7(h) and (i), there is small misclassification in the results of both Nonlocal and ENL-FCN methods when facing Bare Soil class, since the two methods do not make full use of spectral message. The small gap between the ENL-FCN method and the proposed

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TABLE V
CLASSIFICATION ACCURACY (%) AND KAPPA FOR THE IP DATASET

Class	ResNet	ContextNet	SSRN	DPyResNet	A^2S^2K-ResNet	Nonlocal	ENL-FCN	ASPNL
1	94.87±8.17	79.85±8.33	97.98±4.07	**100.00±0.00**	98.81±1.18	99.75±0.73	99.26±1.12	**100.00±0.00**
2	92.88±2.54	96.99±0.68	98.83±0.86	92.71±2.86	**98.99±0.30**	97.32±0.41	97.43±0.44	98.29±0.35
3	89.67±6.07	95.22±1.10	97.85±0.87	87.28±4.89	98.45±0.25	99.52±0.14	99.48±0.30	**99.86±0.14**
4	96.13±5.51	89.60±2.41	97.49±2.54	96.06±2.92	98.39±0.46	97.10±10.7	97.44±1.14	**99.52±0.64**
5	98.61±1.57	96.35±1.93	97.69±1.50	98.47±0.84	98.64±0.05	98.50±0.49	98.87±0.43	**99.11±0.27**
6	97.67±2.42	98.88±0.48	98.01±1.56	95.72±2.78	98.60±0.32	98.77±0.26	98.89±0.42	**99.66±0.17**
7	92.13±9.07	83.53±4.76	96.52±9.04	93.98±9.32	90.47±2.44	**100.00±0.00**	99.60±1.20	**100.00±0.00**
8	94.93±2.82	97.04±1.02	98.65±2.20	95.34±2.48	99.28±0.33	99.46±0.36	99.51±0.28	**99.83±0.11**
9	96.33±7.37	86.69±3.99	96.00±8.00	89.75±15.51	73.89±1.45	70.56±5.58	79.99±8.67	**96.67±5.09**
10	92.47±2.54	97.07±1.17	97.31±1.73	90.36±4.56	97.52±0.61	97.73±0.38	97.56±0.74	**98.74±0.25**
11	94.88±2.60	98.06±0.23	98.84±0.68	94.26±1.75	99.11±0.18	**99.74±0.07**	99.77±0.07	99.66±0.14
12	96.34±1.76	94.56±1.86	97.73±1.60	92.56±4.67	98.25±0.59	98.95±0.34	98.84±0.70	**99.60±0.16**
13	96.16±2.82	97.85±1.60	98.48±1.88	97.91±2.58	99.26±0.44	98.96±0.16	98.52±0.49	**99.34±0.41**
14	95.85±3.00	98.50±0.21	99.37±0.57	96.57±2.33	99.32±0.08	99.89±0.14	99.86±0.15	**99.96±0.06**
15	95.72±2.28	95.15±1.30	98.21±1.05	94.74±2.72	98.24±0.16	95.59±1.37	93.65±2.11	**99.27±0.35**
16	95.39±3.40	95.13±2.88	98.68±2.19	98.34±1.60	97.19±1.76	91.68±2.66	94.09±2.87	**99.04±0.90**
OA	94.42±0.57	96.90±0.64	98.39±0.28	93.63±1.18	98.66±0.33	98.65±0.09	98.65±0.15	**99.38±0.06**
AA	95.00±0.81	93.72±1.54	97.48±1.89	94.62±1.20	96.53±0.90	96.47±0.51	97.05±0.57	**99.28±0.33**
Kappa	93.62±0.65	96.47±0.73	98.16±0.32	92.72±1.35	98.48±0.38	98.47±0.12	98.46±0.17	**99.29±0.07**

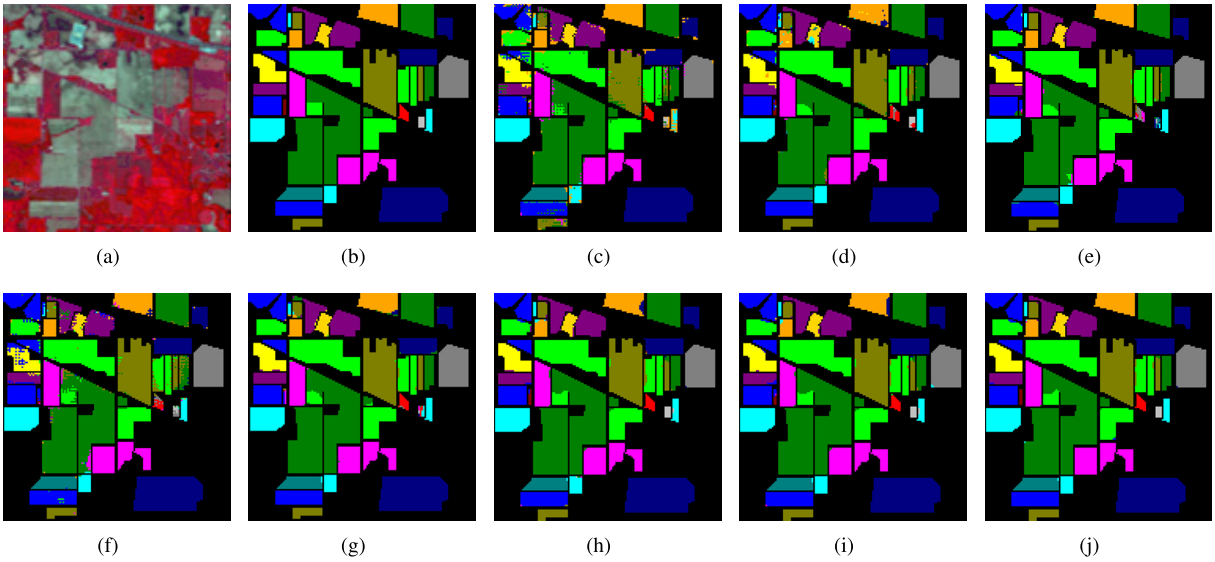


Fig. 5. Classification maps of various algorithms for the IP dataset. (a) False-color composite image. (b) Ground truth. (c) ResNet. (d) ContextNet. (e) SSRN. (f) DPyResNet. (g) A^2S^2K-ResNet. (h) Nonlocal. (i) ENL-FCN. (j) ASPNL.

method shows that the sample imbalance of the UP dataset is not as severe as that of the IP dataset.

D. Ablation Experiment

To further study the effectiveness of the proposed method, this article conducts three groups of ablation experiments. The first is to study the effectiveness of each component, the second is to analyze the impact of feature concatenation on experimental results, and the third is to research the impact of removing the query branch in nonlocal networks.

1) Effectiveness of Components: Table VIII shows the experimental results of the proposed method on three datasets to verify the effectiveness of the three main parts of SNL, APAM, and FNL. In terms of experimental parameter setting, the proportion of the number of training sets used in the experiment on the IP dataset is 10%, and the corresponding ratio on the UP and KSC datasets is 1% and 5%. Also, the OA, AA, and Kappa are used as testing metrics.

The results that SNL, APAM, and FNL are not added are shown in the first row of each part of the three HSI datasets in Table VIII. When the SNL method is added, the experimental results on the three datasets are improved, because long-distance information of the image is taken into account. Taking the IP datasets as an example, OA increases by 0.9%, AA increases by 1.49%, and Kappa increases by

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TABLE VI
CLASSIFICATION ACCURACY (%) AND KAPPA FOR THE KSC DATASET

Class	ResNet	ContextNet	SSRN	DPyResNet	A^2S^2K-ResNet	Nonlocal	ENL-FCN	ASPNL
1	90.66±6.96	98.12±0.77	**100.00±0.00**	93.86±3.90	99.53±0.64	**100.00±0.00**	**100.00±0.00**	99.37±0.63
2	94.97±2.96	94.92±2.92	99.43±0.73	93.96±1.45	98.44±1.49	**100.00±0.00**	99.86±0.27	**100.00±0.00**
3	66.29±17.30	69.31±3.57	96.74±3.42	76.54±2.94	94.53±1.80	98.54±0.53	98.67±0.61	**99.59±0.00**
4	88.46±8.69	74.25±1.30	94.31±6.23	89.52±1.25	95.67±4.75	96.96±1.75	96.41±1.07	**99.79±0.21**
5	83.03±12.35	65.53±1.56	94.97±5.58	81.69±3.99	93.54±1.04	**100.00±0.00**	99.79±0.31	**100.00±0.00**
6	97.98±2.07	88.97±1.26	99.64±0.40	84.02±13.43	99.64±0.28	99.85±0.29	99.85±0.29	**100.00±0.00**
7	96.06±10.72	73.95±2.82	96.43±6.19	96.18±2.05	93.18±2.48	97.34±2.29	97.65±1.83	**100.00±0.00**
8	96.34±2.56	96.77±1.26	98.68±1.53	96.88±0.92	99.30±0.30	99.95±0.14	99.95±0.09	**100.00±0.00**
9	95.12±4.01	98.80±0.48	**100.00±0.00**	97.50±2.13	99.93±0.09	**100.00±0.00**	**100.00±0.00**	99.95±0.08
10	99.66±0.43	99.95±0.07	99.15±1.46	99.39±0.77	99.69±0.36	99.97±0.07	99.92±0.24	**100.00±0.00**
11	**100.00±0.00**	**100.00±0.00**	99.55±0.77	99.93±0.11	99.72±0.60	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
12	92.34±6.72	97.37±0.61	**99.49±0.29**	97.20±1.45	99.24±0.60	96.43±0.18	96.24±0..09	96.23±0.18
13	99.55±0.77	99.95±0.06	**100.00±0.00**	**100.00±0.00**	99.96±0.06	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
OA	94.03±1.11	93.73±0.69	98.67±0.39	94.23±0.92	98.73±0.44	99.37±0.06	99.32±0.03	**99.47±0.11**
AA	92.75±0.92	89.09±1.36	97.87±0.62	92.84±0.37	97.87±0.62	99.15±0.11	99.11±0.06	**99.56±0.09**
Kappa	93.33±1.25	93.02±0.78	98.52±0.44	93.57±1.03	98.59±0.49	99.30±0.07	99.24±0.04	**99.42±0.13**

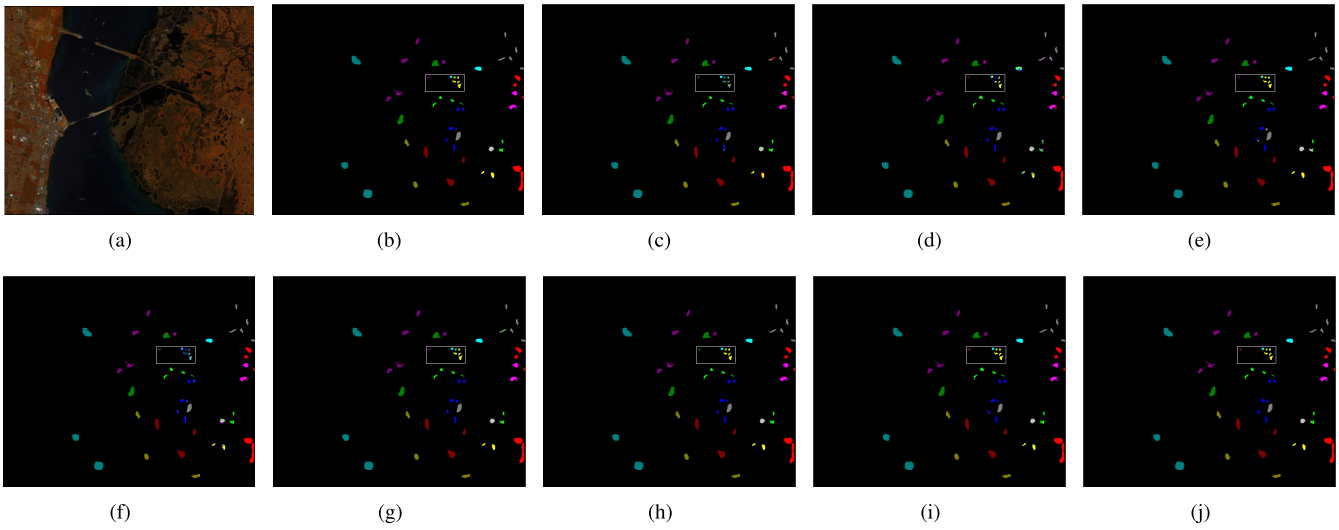


Fig. 6. Classification maps of various algorithms for the KSC dataset. (a) False-color composite image. (b) Ground truth. (c) ResNet. (d) ContextNet. (e) SSRN. (f) DPyResNet. (g) A^2S^2K-ResNet. (h) Nonlocal. (i) ENL-FCN. (j) ASPNL.

1.02%. After adding the SNL and APAM methods, the AA results decreased by 0.16% on the UP dataset, and the other experimental results improved, because the APAM method can better learn the minor samples' information. The best experiment results are achieved when SNL, APAM, and FNL are added. Taking the IP dataset as an example, OA increases by 0.72%, AA increases by 1.54%, and Kappa increases by 0.82%. It is because the multifeature learning can improve the classification results.

2) Influence of Feature Concatenation: ASPNL integrates the outputs of convolutional layer 1 and APAM and takes them as the input of convolutional layer 2. To verify the effectiveness of feature concatenation, experiments are done on three datasets with or without feature concatenation. The experimental outcomes are shown in Table IX.

As shown in Table IX, after adding the feature concatenation, three evaluation indexes OA, AA, and Kappa are improved, which is helpful to the final classification results. Taking the UP dataset as an example, after adding the feature concatenation, OA increases from 98.92% to 99.28%, AA varies from 98.58% to 98.85%, and Kappa changes from 98.58% to 99.06%.

3) Impact of Removing the Query Branch: As aforementioned, the query branch has little effect on the classification results. We conduct experiments on three datasets with or without the query branch to verify this. Table X clearly displays the outcome of the experiment. At the same time, we measured the cosine similarity of the main parts of the nonlocal network on three datasets, and the results are shown in Table XI.

It can be seen from Table X that when adding the query branch to the IP dataset, the experimental results are improved, but the improvement range is less than 0.1%. On the UP and KSC datasets, the experimental results are reduced when

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TABLE VII
CLASSIFICATION ACCURACY (%) AND KAPPA FOR THE UP DATASET

Class	ResNet	ContextNet	SSRN	DPyResNet	A^2S^2K-ResNet	Nonlocal	ENL-FCN	ASPNL
1	63.24±2.11	77.57±2.25	91.72±4.18	66.25±4.19	94.86±0.79	99.16±0.48	**99.38±0.00**	98.99±0.46
2	90.77±2.39	95.02±1.51	97.69±0.75	87.72±2.38	98.08±0.25	**99.99±0.01**	99.98±0.00	99.93±0.07
3	79.03±10.36	53.61±1.94	91.32±5.48	86.56±6.08	89.08±4.33	91.36±2.62	95.45±0.06	**95.87±1.01**
4	98.78±0.93	91.03±2.01	98.89±0.83	97.16±1.52	97.61±2.19	97.17±0.50	96.73±0.03	**97.63±0.19**
5	99.52±0.47	98.07±1.31	99.62±0.24	97.96±2.58	99.35±0.49	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
6	95.87±2.47	91.51±1.76	94.13±3.83	91.01±3.24	96.94±2.18	99.63±0.36	98.33±0.03	**99.91±0.08**
7	88.82±9.79	70.05±3.92	98.10±1.03	80.81±15.55	88.62±7.16	99.29±0.82	97.21±0.06	**99.67±0.26**
8	78.43±9.13	55.72±2.21	85.69±7.03	74.64±8.16	82.33±6.38	98.74±0.46	**99.58±0.00**	98.01±0.49
9	99.89±0.19	78.41±2.12	99.44±0.34	96.91±1.38	97.02±1.87	97.75±0.75	97.59±0.05	**99.81±0.01**
OA	83.63±1.55	85.72±2.03	94.73±0.69	83.31±1.81	94.48±3.21	98.98±0.11	99.07±0.01	**99.28±0.06**
AA	86.33±2.18	79.00±3.28	94.65±0.87	85.75±1.89	93.76±2.90	97.96±0.28	98.29±0.03	**98.85±0.06**
Kappa	77.58±2.29	80.46±2.82	92.98±0.92	77.12±2.69	92.68±4.27	97.65±0.14	98.77±0.02	**99.06±0.07**

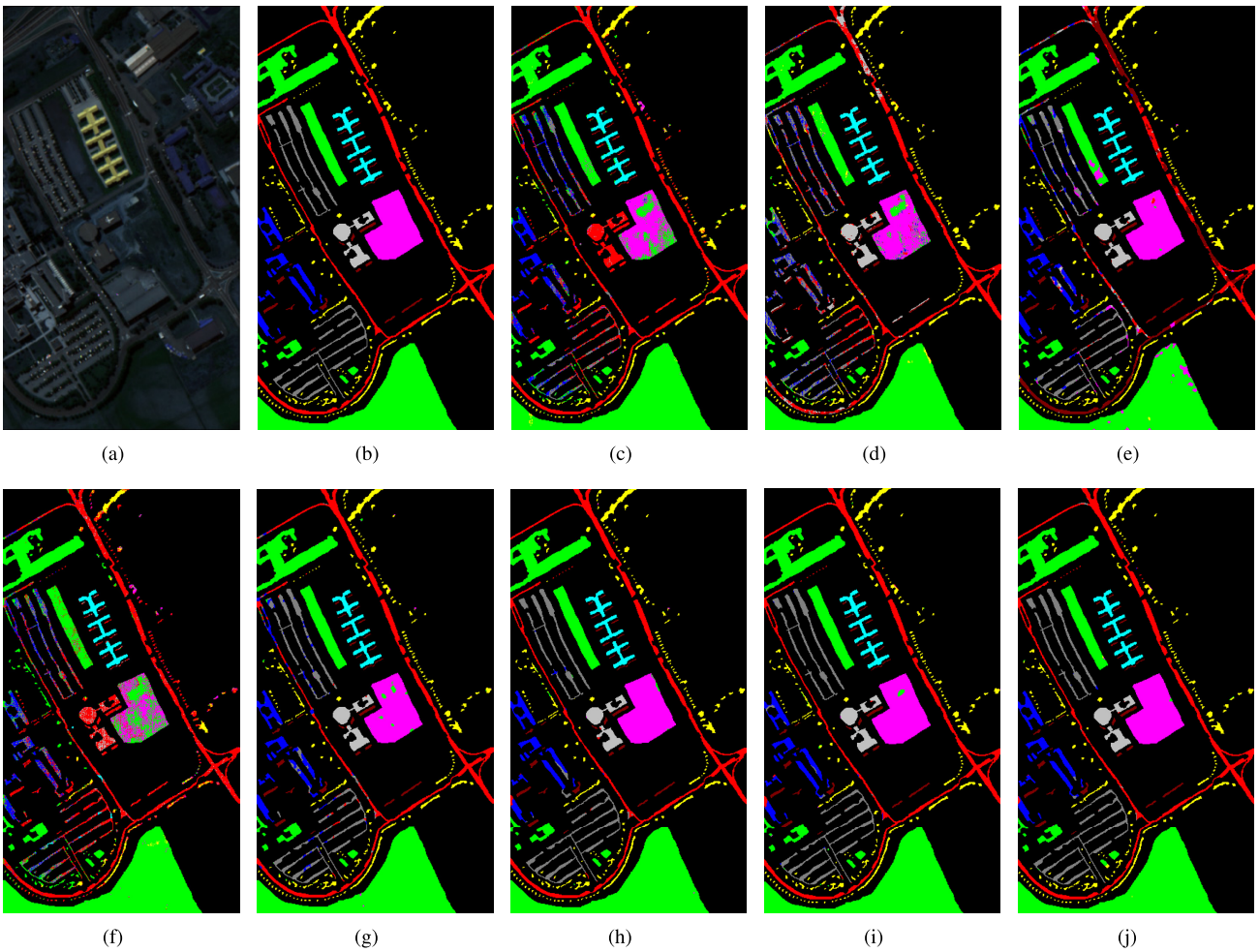


Fig. 7. Classification maps of various algorithms for the UP dataset. (a) False-color composite image. (b) Ground truth. (c) ResNet. (d) ContextNet. (e) SSRN. (f) DPyResNet. (g) A^2S^2K-ResNet. (h) Nonlocal. (i) ENL-FCN. (j) ASPNL.

adding the query branch, but the reduction range is less than 0.1%. Therefore, using a simplified network without the query branch will not cause a significant loss of accuracy.

In Table XI, I represents the input of the nonlocal network, O represents the output, Q represents the query branch, and S represents the similarity matrix calculated by the key branch and the value branch. Each column represents the cosine similarity between different parts of the nonlocal network. Columns 2, 5, and 6 are the cosine similarities among the input of the network, the query branch, the similarity matrix, and the network's output. The cosine similarity between the input and output of the network is the largest, indicating that the output

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TABLE VIII
EXPERIMENTAL RESULTS OF EACH COMPONENT

Dataset	Matrices	OA	AA	Kappa
IP	w/o SNL+APAM+FNL	97.11	95.34	96.71
	w/o SNL+APAM	98.01	96.83	97.73
	w/o SNL	98.66	97.74	98.47
	ASPNL	**99.38**	**99.28**	**99.29**
UP	w/o SNL+APAM+FNL	97.73	94.26	97.01
	w/o SNL+APAM	98.09	97.79	97.49
	w/o SNL	98.63	97.63	97.85
	ASPNL	**99.28**	**98.85**	**99.06**
KSC	w/o SNL+APAM+FNL	98.54	98.64	98.37
	w/o SNL+APAM	99.10	99.01	99.01
	w/o SNL	99.31	99.23	99.23
	ASPNL	**99.47**	**99.56**	**99.42**

TABLE IX
EXPERIMENTAL RESULTS OF FEATURE CONCATENATION

Dataset	Matrices	OA	AA	Kappa
IP	w/o feature concatenation	99.06	98.76	98.94
	w/ feature concatenation	**99.38**	**99.28**	**99.29**
UP	w/o feature concatenation	98.92	98.58	98.58
	w/ feature concatenation	**99.28**	**98.85**	**99.06**
KSC	w/o feature concatenation	99.29	99.36	99.21
	w/ feature concatenation	**99.47**	**99.56**	**99.42**

TABLE X
EXPERIMENTAL RESULTS WITH THE QUERY BRANCH AND WITHOUT THE QUERY BRANCH

Dataset	Matrices	OA	AA	Kappa
IP	w/ Query	**99.43**	**99.32**	**99.35**
	w/o Query	99.38	99.28	99.29
UP	w/ Query	99.22	98.79	99.01
	w/o Query	**99.28**	**98.85**	**99.06**
KSC	w/ Query	99.41	99.52	99.38
	w/o Query	**99.47**	**99.56**	**99.42**

TABLE XI
COSINE SIMILARITY OF MAIN PARTS IN NONLOCAL NETWORK

Dataset	I and O	I and Q	I and S	O and Q	O and S	Q and S
IP	**0.9974**	0.0026	0.5143	**0.0026**	**0.5325**	0.0009
UP	**0.9953**	0.0113	0.8454	**0.0116**	**0.8873**	0.0053
KSC	**0.8679**	0.0212	0.7177	**0.0194**	**0.3718**	0.0074

and the input of the network are the most similar. The cosine similarity between the query branch and the network's output is the smallest, which means that the network's output is the least similar to the query branch. The query branch contributes the least to the output. At the same time, columns 3, 5, and 7 show the cosine similarities between the query branch and the other three parts. These three cosine similarities are relatively small, which means that the similarities between the query branch and the main components of the network are tiny. At the same time, it shows that the query branch contributes little to the calculation of the similarity matrix and the network's final output. In summary, removing the query branch in the nonlocal network is reasonable.

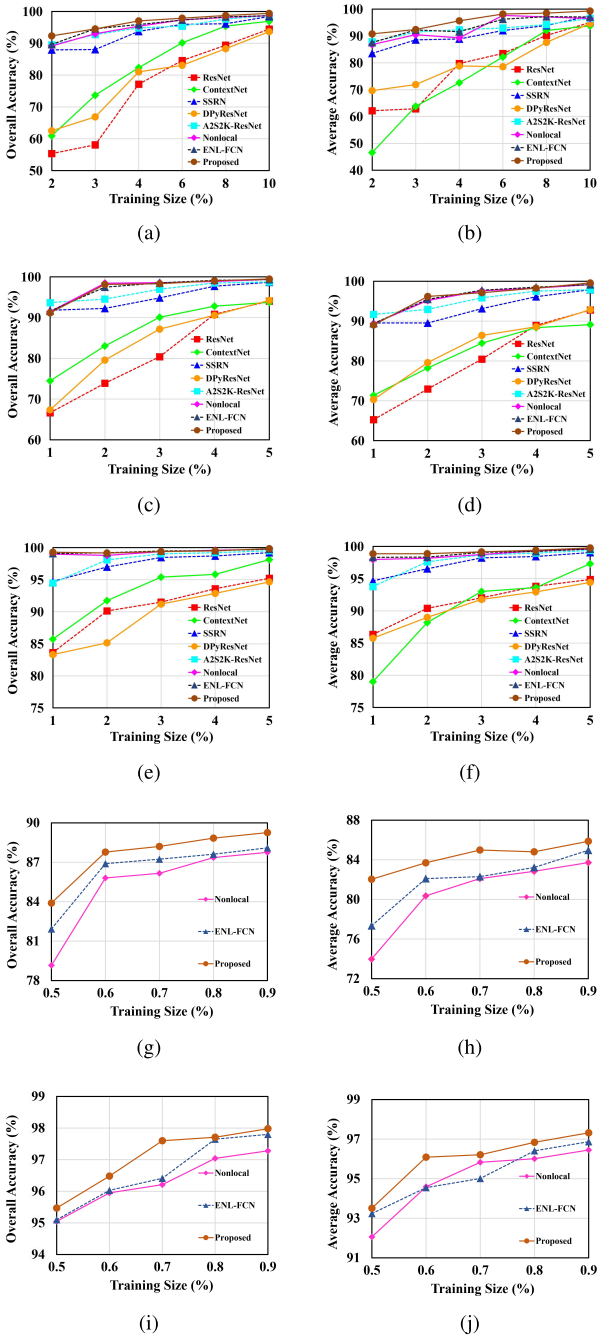


Fig. 8. OA and AA of different training sets. (a) OA and (b) AA on the IP dataset. (c) OA and (d) AA on the KSC dataset. (e) OA and (f) AA on the UP dataset. (g) OA and (h) AA on the KSC dataset with fewer training sets. (i) OA and (j) AA on the UP dataset with fewer training sets.

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E. Influence of Different Numbers of Training Samples

Experiments are designed to examine how different training sets affect the OA and AA of seven comparison methods and the proposed method. Fig. 8 illustrates the results specifically. Fig. 8(a) and (b) shows the OA and AA results on the IP dataset. Fig. 8(c) and (d) shows that on the KSC dataset, the OA and AA of the A^2S^2K-ResNet method achieve the best results when the proportion of the training set is 1%. In other cases, nonlocal-based methods' results are better than other deep learning methods. The consequences on the UP dataset are shown in Fig. 8(e) and (f). It is apparent from the figure that with the increase in the number of training sets, the experimental accuracy of all methods also increases. When the number of training sets is small, the performances of ResNet, Contextnet, and DPyResNet are poor compared with other methods. In most circumstances, the ASPNL produces superior outcomes than other approaches, because ASPNL can better learn the minor samples. In Fig. 8(c) and (e), the results of the Nonlocal, ENL-FCN, and ASPNL are very close, so more experiments are added when the number of training sets is smaller. As shown in Fig. 8(g) and (i), when the training samples are imbalanced due to the small training datasets in the UP and KSC datasets, the OA of the proposed algorithm is higher than those of Nonlocal and ENL-FCN. The results of the corresponding AA are shown in Fig. 8(h) and (j). It shows that the proposed algorithm can learn better than Nonlocal and ENL-FCN with the sample imbalance problem.

IV. Conclusion

The nonlocal neural network has unique advantages in obtaining long-distance semantic information of images. However, its whole image search method easily ignores the knowledge of the minor samples, especially facing with imbalanced HSI. Therefore, this article proposes a simplified nonlocal neural network based on adaptive projection attention for HSI classification. First, this article minimizes the information redundancy and obtains long-distance semantic information of the nonlocal network by using the simplified nonlocal network. Second, this article uses the adaptive projection and compression method to pay attention to changes in a small range and the importance of learning channels to improve the network. Then, by fusing the output results of different layers, feature fusion is realized to improve the robustness of the network. The proposed method is compared with the seven state-of-the-art HSI classification algorithms on three datasets, IP, UP, and KSC. The results show that the proposed algorithm's AA is better than any comparison algorithm, reaching 99.28%, 98.85%, and 99.56%, respectively, on the three datasets. At the same time, for classes with small numbers, the proposed method can always classify more accurately. In the IP dataset with the most imbalanced sample distribution, the proposed method achieves 100%, 100%, and 96.67% accuracy on alfalfa class, grass/pasture moved class, and oats class, respectively. Of course, the proposed method still needs to be improved. For example, selecting information efficiently in SNL will significantly reduce the complexity of the network. At the same time, the experiment results of the UP dataset show that the proposed method has misclassification on some boundaries, so improving the accuracy of boundary classification is also one of our research directions in the future.

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